

Clinical Practice Guideline for an Integrated Approach to Comorbidity in Patients with Psoriasis

Abstract

Background

The relationship between psoriasis and associated diseases has drawn particular interest in recent years. To ensure appropriate management of psoriasis from an early stage, it is essential to promptly diagnose concomitant conditions, prevent the development of comorbidities, and manage them effectively. An integrated approach also ensures that treatments for comorbid conditions do not negatively impact psoriasis control, and vice versa.

Objective

To provide dermatologists with a practical guide focused specifically on the diagnosis and management of the most common comorbidities in patients with psoriasis.

Methods

Diseases included in this guideline were selected based on a systematic literature review. Diagnostic recommendations were derived from established clinical practice guidelines for each condition and validated by a panel of clinical experts. Information on how psoriasis treatments affect comorbid diseases was obtained from drug product characteristics summaries. Data on the impact of comorbidities—and their treatments—on psoriasis were based on a comprehensive review of the scientific literature.

Results

This guideline serves as a concise, user-friendly tool to systematize the

screening, diagnosis, and management of comorbidities in patients with psoriasis, supporting timely referrals and informed treatment decisions.

Conclusion

Implementation of this integrated approach will improve patient outcomes, enhance quality of life, and optimize the use of healthcare resources.

Introduction

The association between psoriasis and various comorbidities has become a major focus of clinical research in recent years. Among these, the link between psoriasis and **psoriatic arthritis (PsA)** is the most well-established, having been recognized for decades. PsA can follow a disease course similar to rheumatoid arthritis and arises from a complex interplay of genetic, immunological, and environmental factors. Genetic studies have identified shared susceptibility loci between psoriasis and PsA, underscoring their pathophysiological overlap.

Other significant comorbidities frequently observed in psoriasis patients include **cardiovascular disease (CVD)** and its major risk factors: **obesity (OB)**, **arterial hypertension (HT)**, **dyslipidaemia (DLP)**, and **type 2 diabetes mellitus (DM)**. These conditions are often clustered under the term **metabolic syndrome (MS)**. The precise mechanism linking psoriasis and CVD remains incompletely understood, but chronic systemic inflammation appears to play a central role. Inflammatory mediators such as **tumor necrosis factor-alpha (TNF- α)**, **interleukins (IL)**, and elevated **C-reactive protein (CRP)** levels—commonly found in psoriasis—also contribute to the development of atherosclerosis and its associated risk factors. The immune pathways involved in psoriasis show considerable overlap with those driving atherosclerotic processes.

Obesity and dyslipidaemia further share pathogenic mechanisms with psoriasis, including adipokine dysregulation and pro-inflammatory cytokine production.

Non-alcoholic fatty liver disease (NAFLD) is another key comorbidity, strongly linked to cardiovascular risk factors and considered by many to be the hepatic manifestation of metabolic syndrome. Emerging evidence indicates a

higher prevalence of NAFLD in individuals with psoriasis compared to the general population.

Inflammatory bowel disease (IBD), encompassing **ulcerative colitis (UC)** and **Crohn's disease (CD)**, also shares genetic and immunological features with psoriasis. Polymorphisms in genes related to the **IL-23 receptor pathway** have been associated with increased risk for psoriasis, UC, and CD, highlighting a common inflammatory axis.

There is also an increased risk of **lymphoma** in patients with psoriasis. This may stem from the underlying immune dysregulation in psoriasis itself or from exposure to systemic therapies such as **cyclosporine** and **methotrexate**, which have been linked to lymphomagenesis. Additionally, patients with psoriasis have a higher incidence of **non-melanoma skin cancer**, influenced by both disease-related immune alterations and long-term immunosuppressive treatment.